CancerOS Implementation Roadmap and Finalist Demo Plan

Document ID: COS-SPEC-012

Version: 1.0

Date: 22 October 2026

Status: Proposed execution backlog and presentation plan

1 Delivery objective and planning assumptions

Deliver a credible breast cancer research prototype that opens a real approved case, renders a measured lesion, navigates observed and hypothetical histories, displays clinical context, accepts a source-supported update and explains verified findings with evidence. The final presentation should make the scientific boundary understandable while demonstrating an integrated working application. Finalist selection is an ambition, not a promised result or known organizer criterion.

This roadmap converts COS-SPEC-007 through COS-SPEC-011 into dependency-ordered coding work. It assumes one primary developer using coding assistance, an 8 GB Ryzen 5 3500U-class laptop, remote heavy preprocessing when needed and no mandatory local GPU training. Exact event deadline, judging rubric, API entitlements and organizer credits/tools remain unverified. Confirm them before committing the presentation duration or sponsor-specific integrations.

Use ten focused working days as a planning example, not an estimate of completed work or a guaranteed schedule. Review effort after the first real case runs end to end. S means roughly half a focused day, M one day, L two or more days; these are initial coding estimates excluding unresolved data access/review delays. A day may contain multiple small tasks; task estimates are not an additive promise of a ten-day total. The milestone schedule requires existing libraries, prepared data and disciplined scope.

2 Priorities and completion definition

Priority	Deliverables	Scope rule
P0 core	Verified baseline, private artifacts, deterministic geometry/measurements, immutable states, charts, five scenarios, updates, evidence-linked summary and genuine real longitudinal case	Required for complete declared prototype
P0 release	Authorization/privacy, critical tests, honest missingness, deployment, recovery and rehearsed demonstration	Cannot be traded for cosmetic polish
P1	Automated segmentation, richer slice viewer, registered overlay, multi-branch comparison, expanded retrieval	Only after mandatory gates pass
Deferred	Whole-body twin, diagnosis/prescribing, treatment ranking, learned patient-specific prediction, production hospital integrations and Kubernetes estate	Do not introduce to rescue the demo narrative

A task is Done only when code is integrated, its meaningful acceptance test passes, contracts/types match, provenance/privacy behavior is checked and the evidence or limitation is recorded. A generated UI button, unverified model response or a screenshot is not task completion. Existing specification files are design inputs; inspect the repository before marking any implementation task complete.

3 Decisions to close before broad coding

Decision	Evidence needed	Default if unresolved
First real baseline and mask	Rights, exact source identity, geometry and annotation meaning	Keep import quarantined; no invented lesion
Genuine follow-up case	Matched original patient, two examinations, comparable measures and valid time domain	Mark P0 gap immediately; continue other work without claiming authentic temporal proof
Inference provider	Account access, configured ID, policy/retention and allowed export	Disable external export; build deterministic summary path
Scenario exposures	Versioned reviewed eligibility rules and known required fields	Use only supported constrained configurations; do not assume receptor values
Contract amendments	Branch/summary discovery, capability/dictionary and evaluation-read shape from COS-SPEC-009	Use bounded approved demo IDs where specified; track production dashboard gap
Event requirements	Official deadline, presentation length, submission format, allowed data/tools and judging rubric	Use adaptable five-minute draft; make no eligibility or sponsor-tool claim

The critical path starts with real data suitability, not model training. Escalate unresolved source rights/linkage by the end of the first focused day. Do not spend most of the schedule polishing a simulator while the required authentic comparison remains unverified.

4 Coding backlog

Dependencies name task IDs. Role labels describe responsibilities, not assumed staff appointments. The primary developer may own engineering, but scientific/source review limitations must remain explicit.

4.1 Foundation and verified vertical slice

ID	Task and output	Depends on	Acceptance and size
B01	Inspect repository, map existing code to specs and create evidence-based gap list	None	Build/start status and unresolved capabilities recorded; S
B02	Resolve selected baseline source, rights and exact mask class; create manifest	None	Reviewed usable source with hashes; M plus access delay
B03	Resolve genuine repeated-imaging case and comparison semantics	B02 or parallel source work	Two real examinations and linkage/time/annotation evidence; L, data-dependent
B04	Establish pinned Python/Next.js environment, minimal CI and secret-free configuration	B01	Reproducible small fixture startup; S
B05	Implement project roles, patient ownership and protected artifact grants	B04	Viewer/editor/admin and cross-patient tests pass; M
B06	Add source/time/clinical/artifact migrations and provenance enforcement	B04, B05	Fresh migration plus invalid-write and immutability tests; L
B07	Implement quarantine import, checksum/metadata validation and explicit publication	B02, B06	Invalid source blocked; approved real case reopens; M
B08	Implement mask measurement, physical transforms, mesh export and QC manifest	B07	Known-array tests and reviewed real overlays pass; L
B09	Build case library, profile, provenance/time labels and private 3D viewer	B05, B07, B08	Actual source lesion rotates with correct units and origin; M

4.2 Longitudinal state and clinical update path

ID	Task and output	Depends on	Acceptance and size
B10	Implement immutable state assembly and explicit member times	B06, B08	Published hash/members stay unchanged after new data; M
B11	Implement compatible numeric comparison and missing/censored semantics	B10	Zero baseline, unit mismatch and definition mismatch handled honestly; M
B12	Build timeline selection, charts/tables and stale-response protection	B09–B11	Rapid switching never mixes state/branch; lab-only view has no fake scan; M
B13	Implement observation batch, correction/retraction, ETags and idempotency	B06, B10	Atomic writes, conflict and preserved-history tests pass; M
B14	Add durable jobs, leases, outbox and selective recomputation	B06, B13	Queue outage/retry/cancel tests; lab change does not regenerate unchanged mesh; L
B15	Integrate actual repeated-imaging case and source comparison report	B03, B10–B12	Genuine visits and comparable measures visible with provenance; M after data approval

4.3 Simulation and intelligence

ID	Task and output	Depends on	Acceptance and size
B16	Implement five versioned scenario definitions and constrained exposure eligibility	B06, B10	Missing required receptor/context blocks incompatible exposure; M
B17	Implement seeded simulator and independent tumor/clinical dimensions	B14, B16	Five families, four timeline positions and repeatable science output; L
B18	Implement generated mask/mesh QC and atomic branch publication	B08, B17	Actual geometry matches target tolerance; failed run invisible; M
B19	Build scenario selection, run status and validation/publication UI	B12, B16–B18	Authorized editor runs/publishes; viewer only reads approved branch; M
B20	Curate minimal rights-reviewed evidence corpus and claim/reference schema	B06	Exact approved passages resolve; patient/background evidence distinct; M
B21	Build deterministic summaries with numeric/source checks	B11, B20	Verified findings, origin and missingness; no invented advice; M
B22	Add optional provider adapter, export policy and structured validation	B14, B20, B21	Export-denial/prompt-injection/grounding tests pass; capability disabled otherwise; M–L
B23	Build findings/evidence inspector and saved/fallback/freshness labels	B12, B21	Each displayed claim resolves correct permitted records; M

B22 is conditional. Evidence-linked interpretation is mandatory; external generation requires its own enabled-provider gate. If a competition explicitly mandates a particular provider, verify that rule and account access early; a fallback does not satisfy an unverified sponsor requirement by assumption.

4.4 Release and presentation work

ID	Task and output	Depends on	Acceptance and size
B24	Close UI API amendments and generate shared clients/contracts	B06, B10, B19, B23	Discovery, dictionaries and report reads documented/tested; M
B25	Run frozen safety/scientific evaluation suite and publish authorized report	B08, B15, B18, B21, B23	Mandatory gates pass; actual counts and failures retained; L
B26	Deploy staging with backups, private objects, health/error monitoring and budget controls	B05, B14	Fresh deployment, role tests and restore verification; M
B27	Benchmark target laptop/browser, mobile/table fallbacks and accessibility tasks	B09, B12, B19, B23	Recorded performance and disclosure/usability findings; M
B28	Create pinned demo manifest and guided navigation with non-destructive reset	B15, B19, B23, B25	All referenced published resources preflight; no privilege bypass; S–M
B29	Prepare final narrative, evidence summary and clearly labeled backup recording	B25–B28	Rehearsed sequence fits verified time; no unsupported capability claims; M
B30	Freeze release, rerun required smoke tests and assemble submission bundle	B24–B29	Commit/deployment/report links aligned, no critical defects or secrets; S

Do not treat B24 as permission to silently alter COS-SPEC-008: revise the versioned contract and generated clients. COS-SPEC-010's synthetic-derived origin rule must be applied consistently before tests and serializers are frozen.

5 Milestones and evidence gates

Milestone	Example working-day window	Required evidence and exit gate
M0 scope and data feasibility	Day 1	Repository gap list, approved baseline candidate, longitudinal candidate and provider decision; no assumed access
M1 real vertical slice	Days 2–3	Authenticated real lesion view with validated mask volume, private grants and physical-space test evidence
M2 actual timeline and updates	Days 4–5	Genuine follow-up comparison, charts/tables and source-backed observation update with immutable history
M3 hypothetical branches	Days 6–7	All five families published with correct shared baseline, generated geometry and repeatability checks
M4 grounded interpretation	Day 8	Evidence inspector, deterministic summary and optional approved inference; clear missingness/failure states
M5 release evidence	Day 9	Frozen evaluation, security/restore/performance results and resolved critical defects
M6 presentation ready	Day 10	Deployed pinned build, preflighted five-minute guide, backup and submission package

Windows overlap and must be re-estimated after M1. If a milestone misses, update the forecast rather than claiming progress from generated code alone. The first cut is optional model/UI complexity; the essential data integrity and true longitudinal evidence remain required. If access/review makes the example schedule infeasible, extend the schedule or explicitly declare the narrower delivered scope.

6 Working approach and hardware strategy

Keep the modular monolith, not a distributed microservice build. Run API and frontend natively; start database/queue services only as memory permits. Use tiny synthetic fixtures for unit work and a small approved real sample for integration. Run heavy image preprocessing remotely where needed, retain immutable outputs and load compact meshes in the browser. Do not make local 3D training or a large local LLM part of the critical path.

Each coding-agent task receives a bounded task ID, input/output contracts, ownership rules and the exact acceptance test. Review diffs before integration, run relevant tests and confirm behavior from the application. Generated code cannot declare a scientific gate passed without its actual inputs and test artifact. Keep repository secrets and protected data out of prompts or fixtures unless the processing environment and permissions allow them.

At the end of each focused work period, record completed task IDs, evidence links, remaining blockers and the next dependency to resolve. Do not measure progress by lines of code or number of screens alone. Save expensive remote processing manifests and source hashes so work can be reproduced without repeating a large download.

7 Scope reduction and contingency decisions

Trigger	Action	What may be claimed
No usable segmentation checkpoint	Use verified imported mask	Correct image-derived reconstruction; no automated segmentation accuracy claim
External provider not permitted/available	Use explicit rule-based or approved saved summary	Grounded summary, with execution mode disclosed
No verified real follow-up case	Continue source search; show requirement unmet	Baseline plus simulation only; not completed authentic temporal proof
Worker slow during presentation	Show actual status and separately open prior completed run	Precomputed validated result; not output from the pending command
Device cannot sustain rich 3D	Reduce visual detail and context layers; maintain measurements	Supported reduced-detail 3D if tested; static fallback is not an interactive 3D pass
Time runs short	Drop optional slice viewer, overlays, forecasting, expanded cohort and advanced retrieval	Delivered P0 capabilities only, with omissions stated
Critical privacy/provenance defect	Disable affected access/output and investigate	Do not present affected feature as ready

The capacity aspiration remains up to 100 baselines and 500 branches; process a smaller approved curated cohort first. For the full demo target, prepare at least five curated entries where feasible, five scenario families from an eligible hero baseline and one genuine longitudinal patient. Reuse a case across purposes transparently; do not inflate distinct real-patient counts.

8 Final presentation sequence

Verify the official presentation duration before finalizing. The default five-minute sequence follows COS-SPEC-009 and can be shortened without relabeling source data. Open a read-only judge view for exploration; live mutations are presenter-only in an isolated authorized demo project.

Time	Screen and action	Suggested factual narration
0:00–0:30	Approved case overview and source summary	CancerOS brings available breast cancer research observations into a time-indexed workspace.
0:30–1:15	Rotate real baseline lesion; open volume/source panel	This geometry comes from the identified mask. The displayed volume is calculated in physical units.
1:15–2:10	Open a named hypothetical branch and move through three follow-ups	The baseline is real. These later states are reproducible simulations, not this person's predicted outcome.

Time	Screen and action	Suggested factual narration
2:10–2:40	Switch to clinical-deterioration scenario	Clinical indicators and lesion burden are separate simulated dimensions. Shrinkage is not equivalent to overall health improvement.
2:40–3:25	Open verified repeated-imaging case and compare two times	These are distinct actual examinations with reviewed linkage and comparable measurement meaning.
3:25–4:10	Add prepared source-supported observation or inspect a prior completed update	New data creates a new revision. Earlier states stay intact, and a laboratory update does not invent a new scan.
4:10–4:45	Open grounded summary and one case/evidence link	Findings link to the selected snapshot and supporting records; this label identifies saved, live or rule-based execution.
4:45–5:00	Show actual evaluation and limitations	These are the measured engineering results and the remaining scientific limits. Patient-specific forecasting is not validated here.

The script's conditional sentences must match actual capability and source evidence. If the genuine repeated-imaging case remains unavailable, state that gap rather than reading its scripted claim. Never keep a percentage or accuracy statement on a slide unless it resolves to a frozen evaluation report.

For a three-minute limit, retain baseline provenance, one simulated trajectory, real follow-up proof and one grounded finding; move live update and detailed evaluation into Q&A. Do not speed through disclosure so the audience mistakes generated data for real history. For an extended session, let judges inspect another family and a failure/abstention case.

9 Rehearsal preflight and fallback

Run preflight on the exact deployed build: authentication, case/source rights, artifact access, snapshot hashes, all required branch/run IDs, known available summaries, real follow-up case, viewer controls and evaluation report. Check source labels at projection size. Confirm no patient/source keys, cloud credentials or private browser content will be exposed by screen sharing.

Rehearse at least three full runs on the presentation laptop/connection, including one deliberate inference outage or slow-job condition. Measure total time and slowest transitions. Keep a clearly labeled backup video and screenshots with the same source disclosures, subject to source rights. A backup is a recording, not a fake live browser session.

Guide reset changes navigation/camera state only. It does not delete user records, cancel jobs or regenerate scenarios. Presenter data reset is a separate controlled administrator action outside the public guide. Do not depend on fresh large uploads, an untested GPU job or an uncertain credit entitlement during the presentation.

10 Submission and evidence package

Provide a README with current scope, local/cloud startup, pinned environments, source access instructions, permissions and limitations; versioned architecture/contracts/safety/evaluation documents; migrations and generated clients; authorized tiny fixtures; test/evaluation reports; deployed judge link with appropriate access; demo guide; screenshots/video; model/data cards; and attribution/license notices. Include reproducibility manifests without protected source maps or secrets.

Avoid redistributing full restricted datasets or checkpoints in the repository. Link to approved acquisition instructions instead. Ensure the submission commit, deployed image, report version and presentation manifest match. Any slide describing planned work must say Planned or Future research. Keep the final repository and deliverables usable after the live session.

11 Final readiness decision

Gate	Full-readiness condition
Data	Approved authentic baseline and genuine matched longitudinal case; correct annotation meaning
Function	Real lesion viewer, synchronized states/charts, five isolated scenarios and immutable update flow
Intelligence	Source-linked validated interpretation, with actual execution mode and honest missingness
Safety	All critical COS-SPEC-010 controls and COS-SPEC-011 mandatory tests pass
Operations	Deployed private artifacts, role enforcement, retry/recovery and restore evidence
Presentation	Timed preflighted guide, actual measured report and accurately labeled backup

The release owner records pass, fail or not_evaluated for each gate and links evidence. Missing required gates mean the full prototype is incomplete; a reduced-scope presentation may still be useful if it clearly states the limitations. No milestone or document alone proves the code is launch-ready or that the project will be selected as a finalist.

12 Related specifications

This backlog depends on the current COS-SPEC-007 architecture, COS-SPEC-008 database/API and COS-SPEC-009 UI/interactive demonstration, together with COS-SPEC-010 safety and COS-SPEC-011 evaluation. Source-data, imaging, simulation and intelligence requirements remain governed by COS-SPEC-003 through COS-SPEC-006. The backlog is ready for implementation planning; estimates, deadlines, completed code and organizer resources must be verified against the actual repository and event information.